



Exercise 1 - Data Science in Astrophysics

DANICA BORTENSTEIN AND VSEVOLOD LANDER

(Dated: March 22, 2026)

Contents

1. Introduction	1
2. Task 1	2
2.1. eCDF of Redshifts	2
2.2. R-Band Magnitude vs Redshift	2
3. Task 2	3
3.1. Colour-Magnitude Diagram	3
4. Task 3	3
4.1. Bootstrapping Statistics and Analytic Confidence Interval	3
5. Task 4	4
5.1. Right Ascension and Declination	4
5.2. Right Ascension and Redshift	4
6. Task 5	4
6.1. Two-Point Correlation	4
7. Task 6	6
7.1. Confidence Interval on Correlation	6
8. Task 7: Discussion	6
9. Appendix	8
9.1. About the uniform distribution on the sphere	8
9.2. About the traps of parallel routines	8

1. INTRODUCTION

By examining the distribution of galaxies in the Universe, we can uncover information about its structure and properties. The *Sloan Digital Sky Survey* (SDSS) is a large survey of astronomical objects, including galaxies. In this report, we will analyze galaxies drawn from SDSS with right ascension in $130 < \alpha/\text{degrees} < 230$ and declination in $5 < \delta/\text{degrees} < 65$. We restrict our redshift range to $0.08 \leq z \leq 0.12$ to minimize the bias resulting from the survey's magnitude limit of $r \approx 17.7$ and sparsity of galaxies in low redshift. Within this range, we classify galaxies by their photometric colours as red ($u - r > 2.3$) and blue ($u - r \leq 2.3$). We then create an angular two-point correlation function $\hat{\xi}(\theta)$ for the two populations using the Landy-Szalay estimator and jackknife resampling to estimate uncertainties on the correlation function. Finally, our correlation function results are compared qualitatively with [Wang et al. \(2013\)](#).

2. TASK 1

2.1. eCDF of Redshifts

In task 1, we were asked to first plot the Empirical Cumulative Distribution Function (eCDF) for the redshift distribution of the catalog, which is given by:

$$\hat{C}(x) = \frac{1}{n} \sum_{i=1}^n \Theta_H(x - X_i) = \frac{\text{number of elements in } S \text{ with } X_i \leq x}{\text{number of elements in } S} \quad (1)$$

Where S is the sample examined that contains elements X_i . For our purposes, this sample was given by the redshifts in the SDSS, and the cumulative fraction is the ratio of the number of galaxies that have a redshift $\leq z$ to the total. As done in the lecture notes, we did this by first ordering the redshifts in ascending order and then plotting the ordered redshift values against the cumulative fraction (Figure 1).

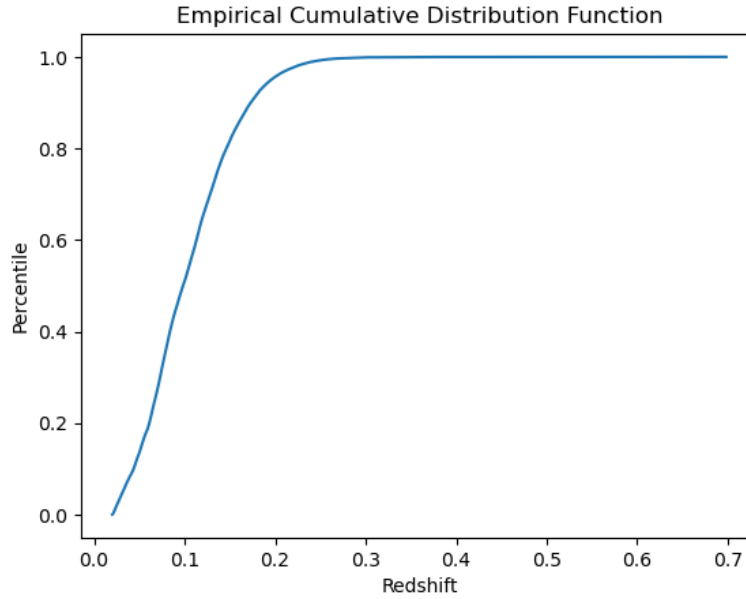
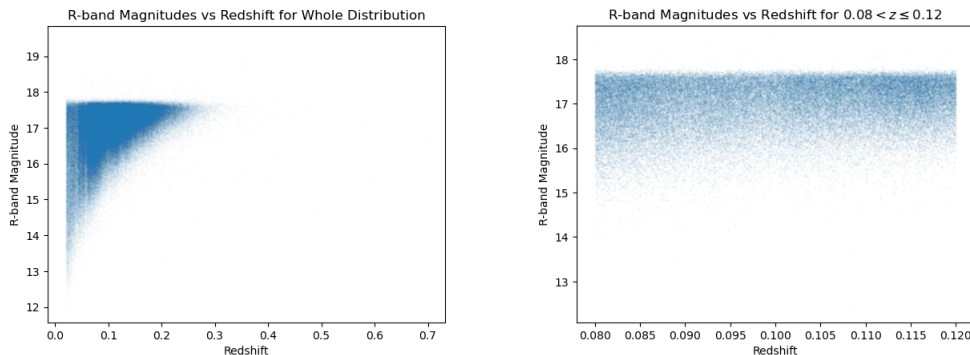


Figure 1: Plot of the empirical cumulative distribution function. The x-value at some height on the y-axis will tell you the redshift below which a certain fraction of the galaxies lie. If the region is steep, more galaxies exist in that redshift range than in a flat region.

2.2. R-Band Magnitude vs Redshift

We were then asked to plot the r-band magnitude against the redshift (Figure 2).



(a) R-band magnitude against redshift for the entire redshift distribution.

(b) R-band magnitude against redshift for $0.08 \leq z \leq 0.12$.

Figure 2: Plots of the r-band magnitude versus redshift for different redshift ranges.

We notice in the scatter plot of the r-band magnitude versus redshift (Figure 2a) that there are two effects: at low z ($\approx z \leq 0.05$), the density of galaxies is relatively sparse but representative of the full magnitude range, and at high z , the volume of galaxies increases but the magnitude limit at $r \approx 17.7$ means that fainter objects are not detected. Selecting a redshift range between 0.08 and 0.12 (Figure 2b) allows for a relatively representative slice where neither of the two effects biases the sample. All further analysis was restricted to this redshift range.

3. TASK 2

3.1. Colour-Magnitude Diagram

Here, we constructed a colour-magnitude scatter plot of the r-band magnitude against the photometric colour ($u-r$) (Figure 3).

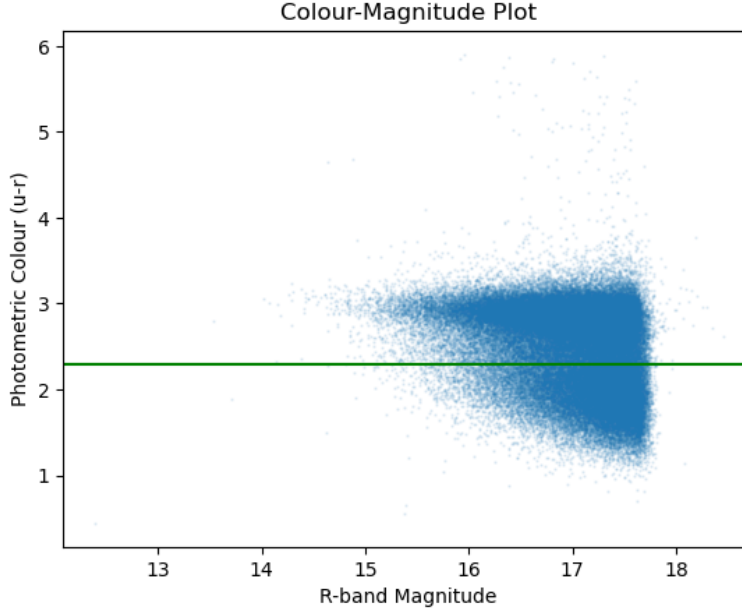


Figure 3: Colour-magnitude plot of $u - r$ against r-band magnitude with a line at $u - r = 2.3$.

We notice in Figure 3 that there are two distinct populations of higher density: one in the range of $u - r \approx 3.5$ to 2.5 and the other in $u - r \approx 1.5$ to 2.2 . These correspond to the red and blue galaxies, respectively. The red galaxies are those with $u - r > 2.3$, for which there are $N = 60656$ galaxies. The blue galaxies are $u - r \leq 2.3$ and there are $N = 30456$.

4. TASK 3

In this section, we first computed the mean and standard deviation of the r-band magnitudes using numpy's `np.mean()` and `np.std()` functions. We found the mean and standard deviation to be 16.89 ± 0.55 for the red galaxies and 17.15 ± 0.43 for the blue galaxies.

4.1. Bootstrapping Statistics and Analytic Confidence Interval

Next we used `scipy.stats.bootstrap()` to compute the 95% confidence interval of the mean magnitude with 1000 resamples. We chose to use the *percentile* method instead of the default *bias-corrected and accelerated (BCa)* method. Primarily, we made this change because our code would not run with the BCa method, but also because the percentile method directly takes the 2.5th and 97.5th percentiles of our resampled means as the confidence interval, which is sufficient given the number of resamples. We chose the percentile method over the *basic* method (which reflects bootstrap errors around the mean) because it does not make assumptions about the symmetry of the bootstrap distribution, making it more robust. After finding the bootstrap confidence intervals, we computed them analytically using the following:

$$\hat{\mu} \pm z_{0.95} \frac{\hat{\sigma}}{\sqrt{n}}, \quad z_{0.95} = 1.96 \quad (2)$$

The results are summarized in (Table 1).

We notice that the 95% confidence interval from the bootstrapping method and the analytic method are in agreement

Galaxy Population	$\hat{\sigma}_{r-band}$	Bootstrapping CI (95%)	Analytic CI (95%)
Red	0.0022	16.8806, 16.8895	16.8805, 16.8893
Blue	0.0026	17.1432, 17.1536	17.1437, 17.1534

Table 1: Summary of the statistics computed by bootstrapping and analytically.

to the 10^{-4} decimal place. The confidence interval for the mean r-band magnitude for the two galaxy populations do not overlap. The red galaxies have a lower mean r-band magnitude than the blue galaxies. Since we took a narrow redshift slice of the data, the galaxies are at approximately the same distance, and so the red galaxies are more luminous on average.

5. TASK 4

5.1. Right Ascension and Declination

In this task, we first plotted right ascension (RA) against declination (DEC) (Figure 4).

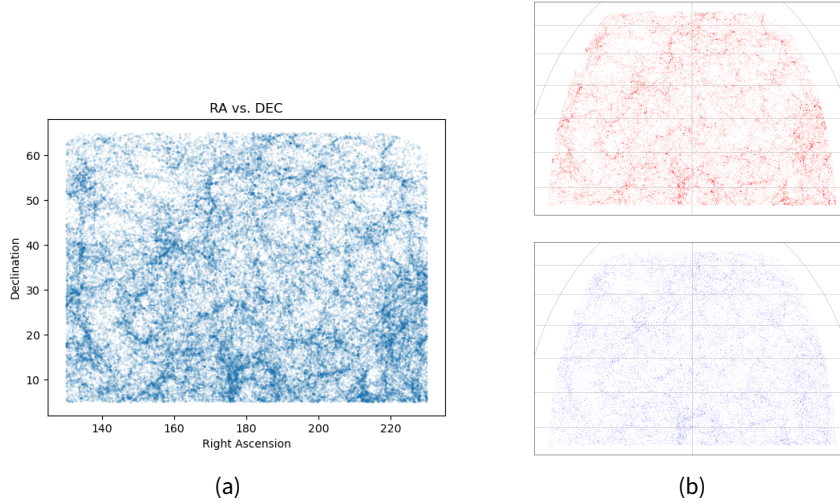


Figure 4: Plots of RA and DEC in different projections: (a) whole distribution, (b) shows the red and blue galaxies using the Mollweide projection.

We can see the structure of the universe in the first plot figure 4a: there are over-dense regions (filaments and clusters) and under-dense regions (voids). In the second plot of galaxies separated by colour, we notice different features for the red and blue galaxies: the red galaxies have more well-defined clustering with clear under-dense regions between, and the blue galaxies appear to be more uniformly distributed with a weaker structure.

5.2. Right Ascension and Redshift

Next we plotted right ascension against redshift (Figure 5)

In the first plot figure 5a, we notice vertical striping in which the clusters appear elongated. This is the so-called "finger of god" effect, where the peculiar velocities of galaxies inside clusters add to their redshift, resulting in the elongating effect in the plot. This effect is also more prominent in the polar plot of the red galaxies in Figure 5b, where we can see radial stretching of filaments. The blue galaxy plot is still more uniform and does not show the finger of god effect as prominently.

6. TASK 5

6.1. Two-Point Correlation

For this task, we were asked to estimate the angular two-point correlation, $\hat{\xi}(\theta)$, function for the red ($\hat{\xi}(\theta)_{red, red}$) and blue ($\hat{\xi}(\theta)_{blue, blue}$) galaxies using the optimized Landy-Szalay estimator which is given by:

$$\hat{\xi}_{LS}(\theta) = \frac{M(M-1) \widehat{DD}(\theta)}{N(N-1) \widehat{RR}(\theta)} - \frac{(M-1) DR(\theta)}{N \widehat{RR}(\theta)} + 1 \quad (3)$$

where N is the number of elements in the dataset, M is the number of points in a random catalog, $\widehat{DD}(\theta)$ is the number of unique data-data pairs with angular separation in bin θ , $DR(\theta)$ is the number of random-data pairs, and

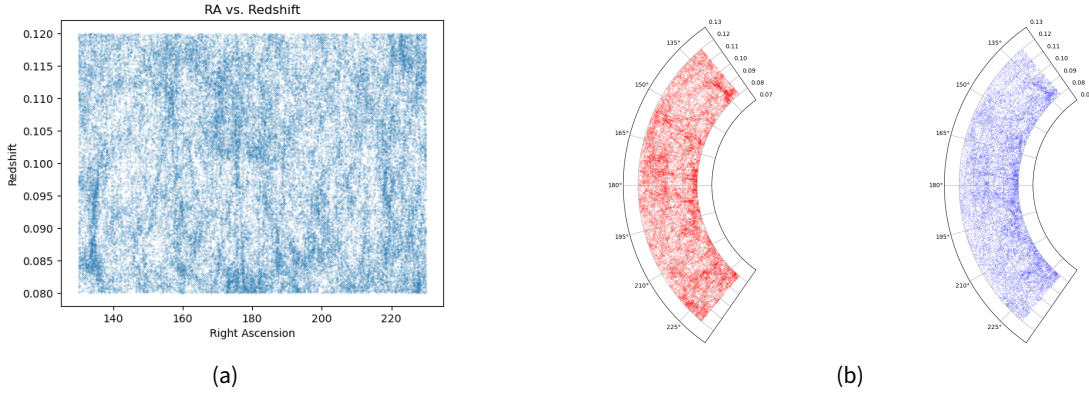


Figure 5: Plots of RA and redshift in different projections: (a) whole distribution in cartesian, (b) shows the red and blue galaxies in the polar map.

$\widehat{RR}(\theta)$ is the number of unique random-random pairs. We first drew $M \approx N^1$ random points from RA in $130^\circ < \alpha < 230^\circ$ and DEC in $5^\circ < \delta < 65^\circ$. We calculated the angular separation with coordinates (α_1, δ_1) and (α_2, δ_2) using a function `dist_sphere` which evaluates the following formula:

$$\omega = \arccos(\sin \delta_1 \sin \delta_2 + \cos \delta_1 \cos \delta_2 \cos(\alpha_1 - \alpha_2)) \quad (4)$$

where this returns the angle between two points on the unit sphere. We then defined two `numba` compiled functions: `spat_spec`, which counts same pairs (DD and RR), and `spat_corr`, which counts DR pairs by iterating over all combinations of the two catalogs. We parallelized using `@jit(parallel=True)` to make looping over the pairs less computationally expensive. The Landy-Szalay estimator `tp_corr_func_LS` was constructed using the resulting pair counts (Figure 6).

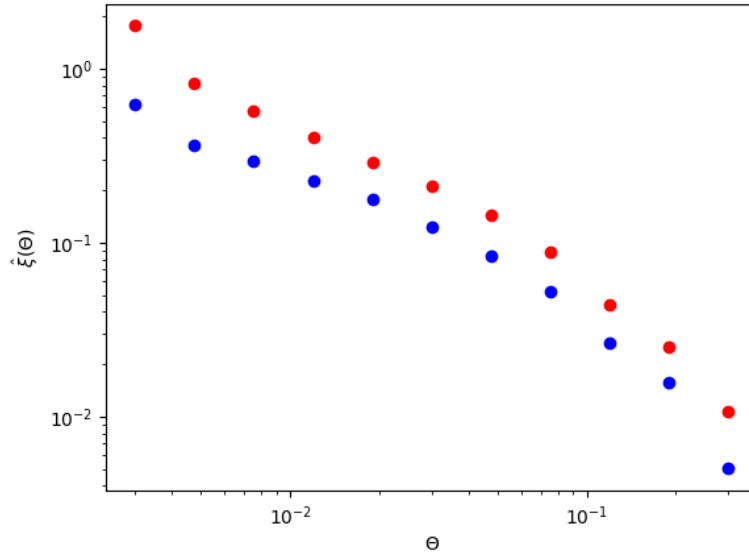


Figure 6: Resulting log-log plot of the correlation function of blue and red galaxies.

The correlation plot is consistent with what we saw in the spatial maps: red galaxies appear to have more pairs at every angular separation compared to a random uniform distribution and a higher $\hat{\xi}(\theta)$ than the blue galaxies (a more uniform spatial distribution). Both galaxy populations show a power-law decline as separation increases, likely due to the clustering weakening at larger scales. The steeper slope for the red galaxies indicates that their clustering is more apparent at smaller scales, and the blue slope being shallower is consistent with their more uniform spatial distribution.

¹ For optimization purposes, we generate random points only once, so we take $M = N_{blue} + N_{red}$. That means $M > N$ for each separate calculation of two-point correlation.

7. TASK 6

7.1. Confidence Interval on Correlation

In task 6, we were asked to add confidence intervals to the correlation functions. Both bootstrapping and ordinary jackknife methods are too expensive in terms of computational time in our case. We decided to use modified delete-d jackknife method. We took N_{it} smaller subsamples and calculated the two-point correlation function $\hat{\xi}_{(i)}$ for each subsample. Then we considered the error of our correlation function from Task 5 to be close to the standard deviation of the correlation functions of each subset. We calculated the mean and standard deviation using the following formulas:

$$\hat{\sigma} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (\hat{\xi}_{(i)} - \bar{\hat{\xi}})^2} \text{ and } \bar{\hat{\xi}} = \frac{1}{n} \sum_{i=1}^n \hat{\xi}_{(i)} \quad (5)$$

We experimented with number and sizes of subsamples and found a balance between accuracy and performance with number of subsamples $N_{it} = 25$ and the number of galaxies in each subsample $N_{sub} = 5000$. The resulting error bars were plotted on the original log-log plot of the two-point correlation function (Figure 7). We observe the error bars

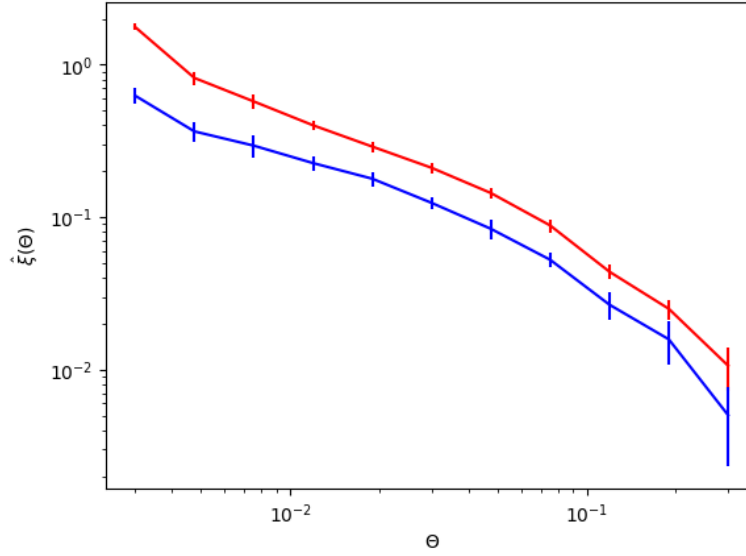


Figure 7: Resulting log-log plot of the correlation function of blue and red galaxies with error bars.

increasing with angular separation for both the red and blue galaxy populations. This could be a consequence of the edge effect: with smaller subsamples, the number of pairs with high separation is dependent on how 'lucky' we are to have galaxies that are close to each other.

8. TASK 7: DISCUSSION

Our results qualitatively agree with those reported in the literature, namely [Wang et al. \(2013\)](#). In figure 8, we see that the two-point correlation function follows a decreasing power law $\omega(\theta) = A_\omega \theta^{1-\gamma}$ with $\gamma = 1.72$ for the full distribution (magnitude $17 < r \leq 21$).

We see that the amplitude of the power law is larger for the red galaxies than for the blue galaxies for all angular separations, given in the right panel of figure 8. This is consistent with our findings from task 5: the red galaxy correlation function lies above that of blue galaxies at all separations, indicating that red galaxies are more clustered and the blue galaxies are more diffuse. They are also consistent with our findings from task 6: error on the correlation function increases with angular separation.

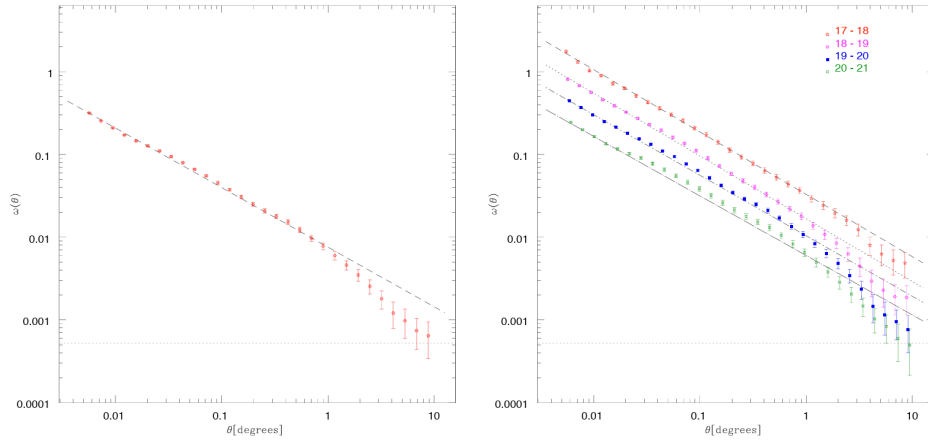


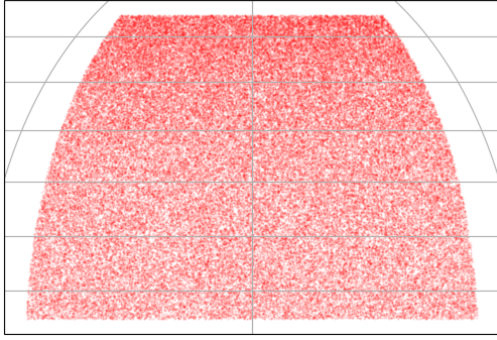
Figure 8: On the left: the two-point angular correlation function for galaxies in the magnitude range $17 < r \leq 21$. Right: the two-point galaxy angular correlation function split by magnitude: notably $17 < r \leq 18$ (red) and $19 < r \leq 20$ (blue) (Wang et al. 2013).

9. APPENDIX

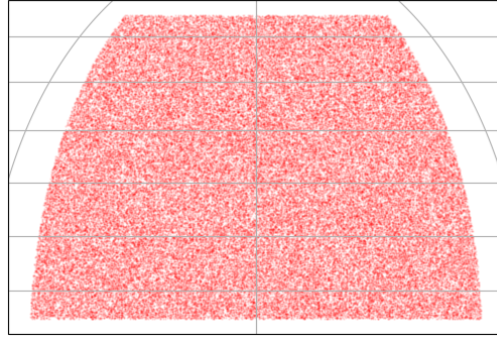
Thoughts in this section are not part of the exercise, but we find them interesting and important for the tasks of this kind. They come from the problems that we met during the work on the code for the two-point correlation function. Since some of the plots in this section are the result of the wrong code and were saved almost by accident, they lack labels and legends. We try to compensate this by more detailed description.

9.1. About the uniform distribution on the sphere

The exercise suggests using a uniform distribution of RA and a uniform distribution of DEC to generate the "uniformly" distributed galaxies on sphere. However in our range of declinations $5^\circ < \delta < 65^\circ$ this will lead to much higher (2 times!) density of galaxies closer to the pole. This will make the resulting correlation function unreliable. So we modified² the generation of the uniform field to make it really uniform on sphere.



(a) RA and DEC are both uniformly distributed: we can see that the resulting field is more dense in the upper part



(b) The correction for the DEC distribution is performed. Now both RA and DEC points are distributed uniformly on the sphere.

Figure 9

9.2. About the traps of parallel routines

At some point of doing this exercise, we met an unusual problem: when we run Landy–Szalay estimator on the whole dataset, it gave us one result, but when we run it on any subsample taken with `np.random.Generator.choice`, the result was completely different, especially at high separations Θ . Even if the subsample had the same length as the original sample, with no element replacement (it's basically the same data in a different order). The reason for this problem was that the pre-compiled parallelized pair-counting function did not work well with 14-core CPU. There were some moments when two sub-processes tried to change the same variable. It was the counter variable, and the simultaneous run of the `n=n+1` operation led to the wrong number of pairs counted in this routine. A simple but slow solution was to turn off the parallelization. Our final solution was to create separate counters for each subroutine and to sum them at the final step of the function.

REFERENCES

Wang, Y., Brunner, R. J., & Dolence, J. C. 2013, Monthly Notices of the Royal Astronomical Society, 432, 1961–1979, doi: [10.1093/mnras/stt450](https://doi.org/10.1093/mnras/stt450)

² Our method is close to this one: [link](#). See more details in our python notebook.